

DuDi: Dual-Scope Discrepancy-Aware Attributed Network Anomaly Detection

Liang Xi , Runze Li, Dichen Li, Ruidong Wang , Han Liu , Huiying Xu , and Xinzhong Zhu , *Member, IEEE*

Abstract—In attributed networks, capturing the correlation between different scopes is crucial to detect anomalies, but few methods focus on modeling the correlation discrepancy of normal and abnormal nodes between different scopes for anomaly detection. Further analysis shows that normal nodes show strong correlation between global and local scopes, while abnormal nodes show weak correlation between the two scopes. Inspired by this perspective, we present a **Dual-scope Discrepancy-aware attributed network anomaly detection method (DuDi)**, to capture macro normal patterns of nodes while reducing noise perturbations based on graph diffusion convolution, and micro patterns of nodes with local graph sampling strategies, respectively, resulting in weak correlation between macro and micro scopes for abnormal nodes, while strong correlation between the two scopes for normal nodes. Based on the dual-scope correlation discrepancy of normal and abnormal nodes, we design a discrepancy-aware anomaly score with Canonical Correlation Analysis (CCA) for anomaly detection. Comprehensive experiments conducted on four real-world datasets showcase the efficacy of DuDi compared to state-of-the-art methods. The experiments show that DuDi achieves significant performance improvements, with up to 6.14% improvement in AUC and 10.33% improvement in AP over the best baselines across multiple datasets.

Index Terms—Attributed network, anomaly detection, correlation discrepancy, graph neural networks.

I. INTRODUCTION

ATTIBUTED network pertains to a network wherein nodes or edges incorporate attribute information. This category of graph data is common in various society networks [1]. Detecting anomalies in attributed networks seeks to identify abnormal nodes, a crucial task extensively applied in various domains [2]. Different from Euclidean data, the unique nature of graph structures introduces additional challenges, specifically in how to appropriately utilize the rich information contained in

the complex “many-to-many” relationships between nodes for detecting anomalous patterns.

Current attributed network anomaly detection approaches can be categorized into two principal paradigms. Graph Representation Learning (GRL) focuses on transforming high-dimensional graph data, including node attributes and topology, into low-dimensional embeddings, through specialized domain-driven techniques, to facilitate anomaly detection in the transformed latent space [3]. Central technologies to this paradigm are Graph Convolutional Networks (GCNs) and Graph Diffusion Convolution (GDC). GCNs aggregate local neighborhood features to model micro-level dependencies [4]. GDC leverages diffusion processes to propagate information across the entire graph while mitigating the noise effects [5]. Although these techniques effectively incorporate both structural and attribute-based information, However, their dependence on smooth feature propagation limits their capacity to capture anomaly patterns, while their general-purpose architectures often lack explicit mechanisms to prioritize features that are crucial for anomaly discrimination [6].

The second paradigm is anomaly detection pattern mining, which directly addresses the identification of anomalous behaviors by developing task-specific mechanisms that quantify deviations from expected patterns. There are mainly three strategies: Reconstruction-based approaches, such as DOMINANT [7], detecting anomalies with reconstruction errors; Self-supervised methods, like CoLA [8], utilizing contrastive learning to distinguish anomalies by maximizing the dissimilarity between corrupted and intact subgraphs; and Anomaly measure methods, which can be classified into one-class-based and community adherence-based approaches.

However, these methods exhibit two major limitations. First, they primarily rely on indirect anomaly proxies (e.g., reconstruction errors, contrastive losses) that inadequately model the inherent multiscale relational discrepancies. More crucially, they fail to reconcile the divergent anomaly signals between macro-level community structures and micro-level ego-network patterns — a critical oversight given recent empirical evidence [9] that normal/abnormal nodes exhibit intrinsically distinct correlation patterns across these scopes: normal nodes maintain strong macro-micro consistency, whereas anomalies manifest pronounced inter-scope discrepancies.

We propose Dual-scope Discrepancy-aware anomaly detection (DuDi; Fig. 1). DuDi amplifies macro–micro discrepancies for anomalous nodes and enforces consistency for normal nodes.

Received 26 March 2025; revised 4 December 2025; accepted 3 February 2026. Date of publication 11 February 2026; date of current version 19 February 2026. This work was supported in part by the Natural Science Foundation of Heilongjiang Province under Grant LH2022F034 and in part by the Chunhui Project Foundation of the Education Department of China under Grant HZKY20220291. The associate editor coordinating the review of this article and approving it for publication was Dr. Dorina Thanou. (*Corresponding author: Liang Xi.*)

Liang Xi, Runze Li, Dichen Li, and Han Liu are with the School of Computer Science and Technology, Harbin University of Science and Technology, Harbin 150080, China (e-mail: xiliang@hrbust.edu.cn; xilaworuo@gmail.com; isjeremylee@gmail.com; mail.liu.han@gmail.com).

Ruidong Wang, Huiying Xu, and Xinzhong Zhu are with the School of Computer Science and Technology, Zhejiang Normal University, Jinhua 321000, China (e-mail: iswangrd@zjnu.edu.cn; xhy@zjnu.edu.cn; zxx@zjnu.edu.cn).

Digital Object Identifier 10.1109/TSIPN.2026.3663969

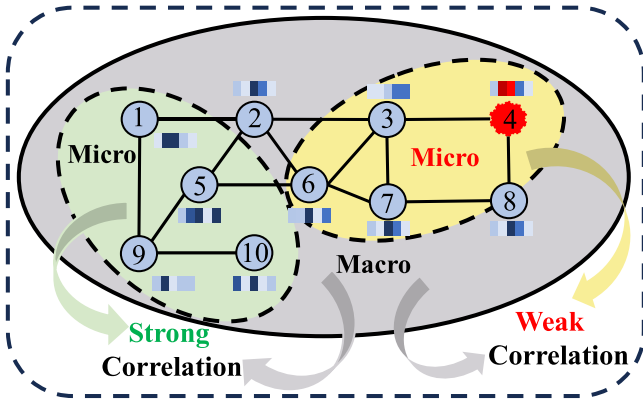


Fig. 1. The motivation of DuDi. For normal node, the correlation between macro and micro scopes is strong; for abnormal node, the correlation between the two scopes is weak.

It establishes a principled detection mechanism based on dual-scope correlation analysis. The main contributions are:

- We propose a dual-scope discrepancy-aware anomaly detection method that detects anomalies by measuring macro–micro correlation discrepancies between weakly correlated abnormal nodes and strongly correlated normal nodes.
- We design a dual-scope correlation discrepancy mining strategy. It accurately extracts micro patterns, captures noise-robust macro normal patterns, and amplifies macro–micro discrepancies for abnormal nodes.
- We conducted various experiments on four real-world datasets, demonstrating that DuDi sets a new state-of-the-art benchmark in attributed network anomaly detection.

II. RELATED WORK

A. Graph Representation Learning

Currently, Graph representation learning methods (GRL) fall into three classifications [3]: matrix factorization, random walks, and graph convolutions.

Matrix factorization methods (e.g., Laplacian Eigenmaps, Graph Factorization) capture global structural patterns [10] but struggle with complex nonlinear structures and scale poorly to large graphs. Random walk–based methods (DeepWalk [11], Node2Vec [12]) learn structure-aware embeddings via simulated walks but overemphasize local context, underrepresent global structure, and ignore node attributes, leading to incomplete representations. Graph convolutional methods such as GCNs [13] and GDC [5] jointly exploit structure and attributes: GCNs aggregate local neighbor features (micro dependencies), whereas GDC diffuses information globally to capture macro dependencies and reduce noise [14]. However, their reliance on smooth propagation can hinder modeling of complex, heterogeneous patterns in graphs [15].

Although these GRL methods can learn local or global patterns, they fail to effectively capture the critical information

required for anomaly detection due to the lack of modeling the discrepancies between these two patterns [6].

B. Graph Anomaly Detection Pattern Mining

To address the limitations of GRL methods for anomaly detection, recent research has focused on specific design for identifying anomalies in graph [9] and dynamic graph data [16]. These approaches integrate structural and attribute representation learning while introducing task-specific mechanisms to effectively capture anomalous patterns. Graph anomaly detection (GAD) methods can be divided into three categories: reconstruction-based methods, self-supervised learning-based methods, and anomaly measure methods.

Reconstruction-based methods operate under the assumption that normal graph instances can be effectively reconstructed, whereas anomalies yield higher reconstruction errors due to structural or attribute irregularities. Representative methods include DOMINANT [7], AnomalyDAE [17], etc.

Self-supervised learning methods, such as CoLA [8], ANEMONE [18], and GRADATE [19], generate data by masking or sampling subgraphs and leverage GNNs to learn embeddings for anomaly detection. Integrating Graph Contrastive Learning (GCL), these methods align similar instances and separate dissimilar ones in embedding space, to improve detection performance [8].

Anomaly measure methods [20] are designed to evaluate the abnormality by incorporating key structural and attribute anomalies into GRL models. Examples of these methods include one-class measures and community adherence measures. One-class-based methods, such as OC-GCN [21] and Dual-SVDAE [22] By evaluating the deviation of samples from a reference distribution, these methods effectively detect anomalies without requiring explicit anomaly annotations. Community adherence-based methods, such as MHGL [23] and GUIDE [24], their assumption is that anomalies show weak or inconsistent community adherence, while normal instances tend to strongly associate with at least one community.

C. Attributed Network Anomaly Detection

Unlike graph anomaly detection, the methods of attributed network anomaly detection not only consider the graph structure but also incorporate node attribute information [25]. CFAD [26] performs anomaly detection by applying pseudo-features to real subgraphs, to generate counterfactual subgraphs. ARISE [27] identifies high-density subgraph structures and performs node attribute learning through contrastive learning, ignoring the macro information of the attributed network. GDAE [28] performs anomaly detection with an autoencoder to simultaneously reconstruct the network structure and node attributes. ADA-GAD [29] denoises nodes, subgraphs, and edges before performing structure and attribute reconstruction. However, due to the non-Euclidean nature of graph data, it is not reasonable for these methods to use the L2 norm as a reconstruction error to measure structural differences.

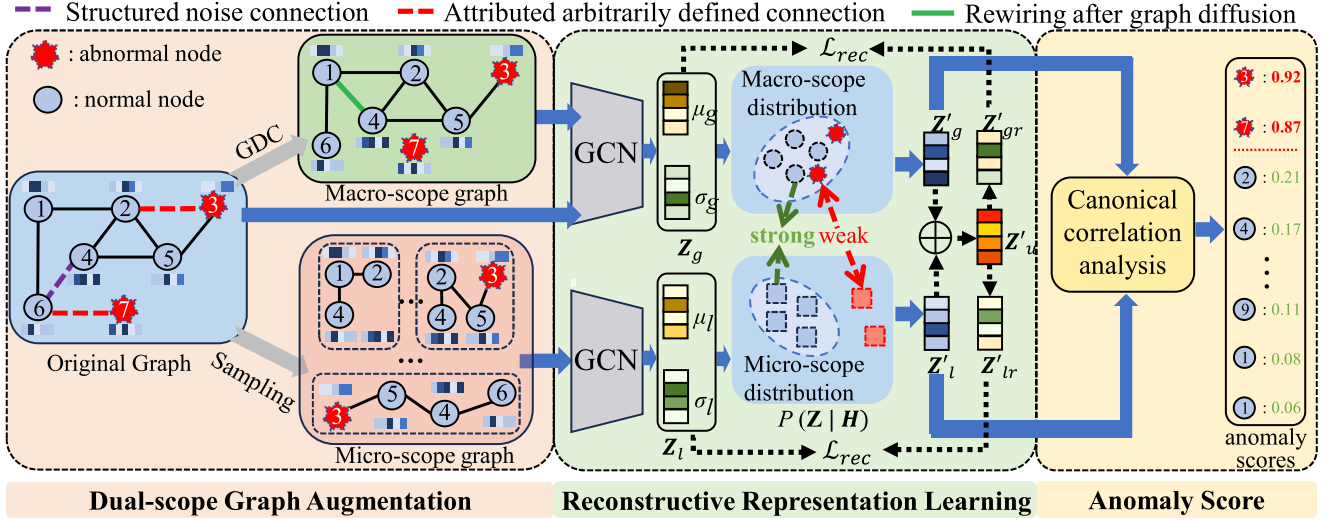


Fig. 2. The framework of DuDi. In the **Dual-scope Graph Augmentation** section, we obtain the Macro-scope and Micro-scope graphs of the original graph using GDC and the random walk algorithm, respectively. In the **Reconstructive Graph Representation Learning** section, we use GCN to extract distribution information and node embeddings for each scope, and apply MLP to reconstruct the node embeddings. During training, the distributions of the two scopes are aligned, and CCA is used to model the discrepancy between the two scopes. Finally, the Canonical Correlation Analysis is employed as the anomaly score for **anomaly detection**.

Existing methods primarily perform indirect anomaly detection via model inference and lack explicit anomaly pattern modeling mechanisms [30]. They usually operate at a single macro or micro level, overlooking correlation discrepancies across scopes [31], motivating an explicit anomaly pattern recognition framework that exploits macro–micro correlation differences between normal and abnormal nodes in attributed networks.

III. METHODOLOGY

As shown in Fig. 2, DuDi consists of three parts: dual-scope graph augmentation, reconstructive graph representation learning, and discrepancy-aware anomaly score computation. (1) We apply GDC to the original graph to obtain a rewired diffusion graph as the macro scope, and use subgraph sampling around each node to derive subgraph representation sequences as micro scopes. (2) A shared-weight GCN encodes the original and diffusion graphs as macro-scope embeddings, while another GCN encodes subgraph sequences as micro-scope embeddings; KL divergence aligns the two distributions, and the fused features are used for reconstruction learning. (3) Based on Canonical Correlation Analysis (CCA), we formulate the anomaly score of each node as the correlation discrepancy between macro scope and micro scope, with the nodes exhibiting weak correlation identified as anomalous nodes.

A. Dual-Scope Graph Augmentation

It consists of two components: macro graph augmentation with GDC, and micro graph augmentation by sub-graph sampling, for increasing the correlation consistency of normal nodes across global and local scopes and amplifying the correlation discrepancy of anomalies between the two scopes.

1) *Macro Graph Augmentation*: Our primary objective is to enhance the information propagation across the entire network to

effectively mitigate noise and diminish arbitrary structural connections. graph diffusion can leverage the inherent connectivity of the graph to distribute information or features, improving interactions and integration among nodes [14]. Graph Diffusion Convolution (GDC) introduces a denoising mechanism through sparse graph diffusion topology, which strategically suppresses high-frequency noise in local neighborhoods (e.g., spurious edges or transient attribute fluctuations). By propagating information over multi-hop diffusion paths with exponentially decaying weights, GDC effectively filters out localized interference while amplifying consistent interaction patterns across the graph, thereby enhancing the fidelity of global normal pattern extraction in anomaly detection tasks.

Given an undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ with node set $\mathcal{V}$ and edge set $\mathcal{E}$. $\mathcal{N} = |\mathcal{V}|$ is the number of nodes and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the adjacency matrix. We define generalized graph diffusion via the diffusion matrix with the weighting coefficients θ_k and the generalized transition matrix $\mathbf{T}$.

$$\mathbf{S} = \sum_{k=0}^{\infty} \theta_k \mathbf{T}^k. \quad (1)$$

The choices for θ_k and $\mathbf{T}^k$ can be quite diverse, yet they must satisfy the condition that (1) converges [14]. Some stricter conditions require that $\sum_{k=0}^{\infty} \theta_k = 1$, in which $\theta_k \in [0, 1]$, and the eigenvalues of $\mathbf{T}$ denoted as λ_i must fall within the range $[0, 1]$ to ensure convergence. For the transition matrix, the following three types are commonly used: random walk transition matrix $\mathbf{T}_{rw} = \mathbf{A}\mathbf{D}^{-1}$, symmetric transition matrix $\mathbf{T}_{sym} = \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$ and adjacency matrix with added weighted self-loops $\tilde{\mathbf{T}}_{sym} = (\omega_{loop}\mathbf{I}_N + \mathbf{D})^{-\frac{1}{2}}(\omega_{loop}\mathbf{I}_N + \mathbf{A})(\omega_{loop}\mathbf{I}_N + \mathbf{D})^{-\frac{1}{2}}$.

Among the diverse options, two popular examples of graph diffusion are personalized PageRank (PPR) and the heat kernel.

In the context of PPR, the choice is made such that $\mathbf{T} = \mathbf{T}_{rw}$ and the parameter $\theta_k^{PPR} = \alpha(1 - \alpha)$, where the teleport probability α lies in the interval $[0, 1]$. For the heat kernel, it is defined with $\mathbf{T} = \mathbf{T}_{rw}$ and $\theta_k^{HK} = e^{-t \frac{t_k}{k!}}$, where t represents the diffusion time. In DuDi, we opt for PPR-based graph diffusion to assist in graph learning, as both the features and edges in actual graphs frequently exhibit noise. In this way, we can capture macro normal patterns of nodes with effective avoidance of noise perturbations.

The construction of the macro graph is achieved by removing attribute-defined connections. The primary motivation is to minimize the impact of noise and irrelevant connections that might be introduced by attribute-based links. These attribute-defined connections often add unnecessary complexity to the graph, potentially distorting the true structural relationships between nodes. Eliminating such connections can ensure that the graph structure more accurately reflects the actual relationships among nodes, to enhance the efficiency of information propagation during the graph diffusion process and the fidelity of global pattern extraction in anomaly detection tasks.

To sum up, the original graph $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$ is fed into the PPR-based graph diffusion module to rewire and generate a diffused graph $\mathcal{G}_{macro} = \{\mathbf{A}_g, \mathbf{X}_g\}$, from a macro perspective to smooth the perturbations caused by structural connection noise and arbitrarily defined attribute connections, eliminate the restriction of using only direct neighbors, recover meaningful neighborhoods from noisy graphs, and capture macro normal patterns of nodes, eventually.

2) *Micro Graph Augmentation*: We focus on utilizing the local connectivity information to deeply explore the immediate environment of the nodes and their micro topological structures [32]. Building on the work of [33], we adopt a sub-graph sampling strategy based on random walk, to explore the graph by sequentially visiting neighboring nodes, effectively capturing and maintaining the micro topological structure, connectivity, and contextual relationships between nodes.

As a result, the random walk algorithm is employed on $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$ to generate a sub-graph sampling sequence $\mathcal{G}_{micro} = \{\mathbf{A}_l, \mathbf{X}_l\}$, from a micro perspective, to effectively capture the connectivity patterns and features within a small-scale range for each node, which leads to a better reflection on the centrality and similarity on the micro scope of nodes.

B. Reconstructive Graph Representation Learning

This module is to capture the distribution of macro-scope and micro-scope node attributes.

1) *Dual-Scope Encoding*: For $\mathcal{G}_{macro}$, because the diffusion process can blur the relationships between nodes, leading to the diminished visibility of subtle features or fine differences among nodes in the original graph, we use a weight-sharing GCN, jointly the attribute information of $\mathcal{G}$ and structural information of $\mathcal{G}_{macro}$, to exploit their shared macro structural similarities and ensure efficient and consistent macro-scope feature fusion that focuses on the distant connections of nodes and the influence of neighbors, simultaneously.

$$\text{GCN}(\mathbf{H}, \mathbf{A}|\mathbf{W}) = \varphi((\mathbf{D})^{-\frac{1}{2}} \mathbf{A}(\mathbf{D})^{-\frac{1}{2}} \mathbf{H}\mathbf{W}), \quad (2)$$

where $\mathbf{D}$ is the diagonal degree matrix of $\mathcal{G}_{macro}$, $\mathbf{H} = \mathbf{X}$ is the node attributes of $\mathcal{G}$, $\mathbf{A}$ is the adjacency matrix of $\mathcal{G}_{macro}$.

For $\mathcal{G}_{micro}$, as micro structures often exhibit greater heterogeneity than macro ones, we employ a dedicated aggregation method to extract the aggregated micro-scope embeddings of nodes. Avoiding weight sharing with the former GCN allows for more precise capture of micro variations, enhancing the DuDi's ability to capture anomalous patterns. We input $\mathcal{G}_{micro}$ to the encoder:

$$\text{GCN}(\mathbf{X}, \mathbf{I}|\mathbf{W}) = \varphi(\mathbf{I}\mathbf{X}\mathbf{W}), \quad (3)$$

where $\mathbf{I}$ is the sub-graph sampling sequence from $\mathcal{G}_{micro}$, which, while retaining more fine-grained information, particularly focusing on detailed node features at a small scale, transfers the local context information of nodes from the random walk sequence to micro node features. $\mathbf{W}$ in (2) and (3) is the trainable weight. φ is the activation function, such as $\text{RELU}(\bullet) = \max(0, \bullet)$.

2) *Distribution Alignment and Decoding*: After dual-scope GCN, the macro-scope and micro-scope distributions are learned by (4), respectively.

$$q(\mathbf{Z}|\mathbf{X}, \mathbf{A}) = \prod_{i=0}^N q(\mathbf{z}_i|\mathbf{X}, \mathbf{A}), \quad (4)$$

$$q(\mathbf{z}_i|\mathbf{X}, \mathbf{A}) = \mathcal{N}(\mathbf{z}_i|\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}^2)), \quad (5)$$

where $\boldsymbol{\mu}$ denotes the vectors of mean values, $\boldsymbol{\sigma}$ denotes the vectors of variances. The embeddings, represented by $\mathbf{Z}$, are derived from the dual-scope distributions. In terms of the macro-scope distribution, $\boldsymbol{\mu}_g$ and $\boldsymbol{\sigma}_g$ stand for the mean and variance vectors, respectively, while $\mathbf{Z}_g$ represents the macro-scope embeddings. Similarly, we can get $\boldsymbol{\mu}_l$, $\boldsymbol{\sigma}_l$ and $\mathbf{Z}_l$.

$$\boldsymbol{\mu} = \text{GCN}_{\boldsymbol{\mu}}(\mathbf{H}, \mathbf{A}|\mathbf{W}), \quad (6)$$

$$\boldsymbol{\sigma} = \text{GCN}_{\boldsymbol{\sigma}}(\mathbf{H}, \mathbf{A}|\mathbf{W}). \quad (7)$$

Given the complexity of directly aligning these two distributions, we adopt the Kullback-Leibler (KL) divergence to pull the macro distribution $q(\mathbf{Z}_g|\mathbf{X}_g, \mathbf{A}_g)$ and the micro distribution $q(\mathbf{Z}_l|\mathbf{X}_l, \mathbf{A}_l)$ towards a Gaussian distribution, which serves as the prior [34], [35].

$$\begin{aligned} \mathcal{L}_{kl} = & -\text{KL}[q(\mathbf{Z}_g|\mathbf{X}_g, \mathbf{A}_g)||p(\mathbf{Z}_g)] - \text{KL}[q(\mathbf{Z}_l|\mathbf{X}_l, \mathbf{A}_l)||p(\mathbf{Z}_l)], \end{aligned} \quad (8)$$

where $p(\mathbf{Z}_g)$ and $p(\mathbf{Z}_l)$ represent the prior Gaussian distributions for the macro and micro scopes, respectively. The KL divergence quantifies the difference between the learned distributions and the prior, encouraging the model to pull both distributions towards a common prior Gaussian distribution. This approach effectively aligns the two distributions to a shared latent space, ensuring consistency in the embedding space, and enabling the subsequent reconstruction process to effectively utilize both macro and micro-level information.

After distribution aligning, we normalize the node embeddings $\mathbf{Z}_g$ and $\mathbf{Z}_l$ from these two scopes as follows:

$$\begin{aligned}\mathbf{Z}'_g &= \frac{\mathbf{Z}_g - \mu(\mathbf{Z}_g)}{\sigma(\mathbf{Z}_g) * N^{\frac{1}{2}}}, \\ \mathbf{Z}'_l &= \frac{\mathbf{Z}_l - \mu(\mathbf{Z}_l)}{\sigma(\mathbf{Z}_l) * N^{\frac{1}{2}}},\end{aligned}\quad (9)$$

we design a distribution reconstruction decoder based on Multi-Layer Perceptron (MLP) for reconstructing features from the two scopes, $\mathbf{Z}'_{gr}$ and $\mathbf{Z}'_{lr}$, based on the fused embeddings, $\mathbf{Z}'_u$.

C. Discrepancy-Aware Anomaly Score

This discrepancy-aware anomaly score of DuDi is dedicated to detecting the correlation discrepancy between macro-scope and micro-scope embedding distributions. We employ Canonical Correlation Analysis (CCA) [36] to explicitly model the discrepancy between the two scopes.

DuDi aims to maximize the correlation between the two scope distributions by minimizing the invariance between $\mathbf{Z}_g$ and $\mathbf{Z}_l$.

$$\mathcal{L}_{inv} = \|\mathbf{Z}'_g - \mathbf{Z}'_l\|_F^2. \quad (10)$$

To prevent the method from converging to trivial solutions, we introduce $\mathcal{L}_{dco}$ loss to ensure that the individual dimensions of the features are uncorrelated:

$$\mathcal{L}_{dco} = \|\mathbf{Z}'_g{}^T \mathbf{Z}'_g - \mathbf{I}\|_F^2 + \|\mathbf{Z}'_l{}^T \mathbf{Z}'_l - \mathbf{I}\|_F^2. \quad (11)$$

The objective based on CCA is defined as follows, in which λ is the trade-off between two terms:

$$\mathcal{L}_{CCA} = \mathcal{L}_{inv} + \lambda * \mathcal{L}_{dco}. \quad (12)$$

$\mathcal{L}_{CCA}$ enhances sensitivity to anomalous patterns by explicitly modeling the correlation discrepancy between the macro-scope and micro-scope distributions. By minimizing the invariance loss $\mathcal{L}_{inv}$, it ensures the alignment of embeddings from both perspectives, further improving sensitivity to anomalies. The introduction of $\mathcal{L}_{dco}$ ensures the independence of feature dimensions, preventing the potential redundancy and feature dependency issues that may arise during alignment. This approach avoids solely relying on reconstruction error, enabling the model to more effectively capture subtle discrepancies between the macro and micro distributions.

The total loss of DuDi is composed of three parts: reconstruction loss, KL divergence loss, and CCA loss:

$$\mathcal{L} = \mathcal{L}_{Rec} + \mathcal{L}_{CCA} + \mathcal{L}_{KL}. \quad (13)$$

Differing from other approaches, the anomaly score of DuDi is characterized by the correlation between two scopes. A high correlation between the global-scope embeddings and the local-scope embeddings is deemed normal, while a low correlation is considered abnormal.

Algorithm 1: DuDi.

Require:

Attributed network $\mathcal{G} = \{\mathbf{A}, \mathbf{X}\}$, encoder f_g and f_l , the prior distribution p , and the hyper-parameter λ of of CCA.

- 1: Generate global graph $\mathcal{G}_{macro} = \{\mathbf{A}_g, \mathbf{X}_g\}$ via (1) and sample local sub-graphs $\mathcal{G}_{micro} = \{\mathbf{A}_l, \mathbf{X}_l\}$ by random walk.
 - 2: **for** $epoch = 1$ to T **do**
 - 3: Generate the macro graph embedding distribution $q(\mathbf{Z}_g|\mathbf{X}_g, \mathbf{A}_g)$ and the micro graph embedding distribution $q(\mathbf{Z}_l|\mathbf{X}_l, \mathbf{A}_l)$ via (2) and (3).
 - 4: Align the two distributions via (8).
 - 5: Sample macro-scope graph embeddings $\mathbf{Z}_g$ and micro-scope graph embeddings $\mathbf{Z}_l$ from the two distribution.
 - 6: Normalize $\mathbf{Z}_g$ and $\mathbf{Z}_l$ by (9).
 - 7: Input the normalized $\mathbf{Z}'_g$ and $\mathbf{Z}'_l$ into CCA, and update DuDi with its stochastic gradient by (13).
 - 8: **end for**
 - 9: Compute the correlation between $\mathbf{Z}'_g$ and $\mathbf{Z}'_l$ via (10) and detect the anomaly by it.
 - 10: **return** encoder f_g and f_l .
-

TABLE I
STATISTICS OF THE USED REAL-WORLD DATASETS

Dataset	$\mathcal{V}$	$\mathcal{E}$	$\mathcal{A}$	Class
Cora	2, 708	5, 429	1, 433	7
Citeseer	3, 327	4, 732	3, 703	6
Pubmed	19, 717	44, 338	500	3
Bitotc	4, 863	40,983	128	2

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets

We select four publicly available datasets: Cora, Citeseer, Pubmed¹ and Bitotc [37] for experiments. The statistics are provided in Table I. We designated one class as the normal class and grouped the remaining classes as the abnormal class for each dataset. We randomly sample 60% of normal nodes for training, 15% and 25% of normal nodes and the same number of abnormal nodes for validation and testing, respectively [38].

B. Baselines

To evaluate the effectiveness of DuDi, we select four different types of baselines: Traditional one-class-based methods, such as OC-SVM [39] and Deep-SVDD [40], which employ graph embedding techniques (e.g. DeepWalk [11]) to represent nodes in a latent space using structural information, followed by one-class classification for anomaly detection. Graph autoencoder based methods, including Dominant [7] and Conad [41], which identify anomalies by ranking the reconstruction errors.

¹<http://linqs.cs.umd.edu/projects/projects/lbc>

TABLE II
ANOMALY DETECTION PERFORMANCE. ALL VALUES ARE PERCENTAGES (%) AND AVERAGED WITH STDDEVS OVER 5 SEEDS. THE BEST RESULTS ARE MARKED IN **BOLD** AND THE SECOND-BEST RESULTS RESULTS IN *ITALIC*.

Dataset	Cora		Citeseer		Pubmed		Bitotc	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
OC-SVM(Raw) [39]	60.92±5.41	57.51±4.48	55.42±3.79	54.74±2.67	54.62±9.67	53.89±7.16	24.10±0.88	12.11±0.23
OC-SVM(DW) [39]	66.83±12.82	64.73±11.86	50.37±12.68	51.84±8.12	52.40±8.40	52.35±6.16	23.71±3.65	13.51±2.87
Deep-SVDD(Attr) [40]	63.81±3.32	62.21±3.50	67.18±7.33	66.73±7.82	70.77±9.44	71.03±10.64	73.29±0.72	63.21±1.60
Deep-SVDD(Stru) [40]	70.13±4.51	69.21±5.30	73.47±10.23	73.70±9.74	71.39±9.77	71.62±11.02	75.89±2.14	56.21±7.97
Dominant [7]	87.29±3.71	78.35±6.23	80.17±7.78	72.36±6.15	65.29±2.21	61.66±2.47	38.68±5.46	18.05±2.45
Conad [41]	52.63±2.44	53.73±2.82	52.80±3.82	54.42±4.61	53.54±5.53	53.27±4.36	45.89±6.72	18.92±3.00
OC-GCN [21]	68.19±11.19	66.76±9.15	75.29±6.55	73.29±6.35	66.65±5.37	68.05±6.22	75.46±2.09	54.41±7.71
OC-SAGE [21]	90.99±2.60	89.37±3.12	84.13±6.67	81.52±5.53	74.60±6.17	75.11±6.83	67.78±2.83	34.80±4.81
CCA-SSG [42]	89.40±5.78	82.01±9.35	81.86±7.69	73.58±7.00	69.83±11.49	62.58±8.45	66.63±1.72	27.63±1.66
OC-GRADATE [19]	50.86±6.60	86.03±7.76	51.33±6.05	81.82±4.69	51.08±1.19	66.77±9.75	46.49±1.10	59.10±0.03
OC-GGAD [43]	94.40±3.15	88.70±7.79	83.99±8.25	76.74±8.69	82.12±1.82	74.31±3.04	81.74±1.26	48.73±1.83
DuDi	94.23±3.24	93.04±3.99	85.11±6.18	82.71±7.11	84.85±3.13	83.52±3.71	80.09±1.04	60.51±1.69

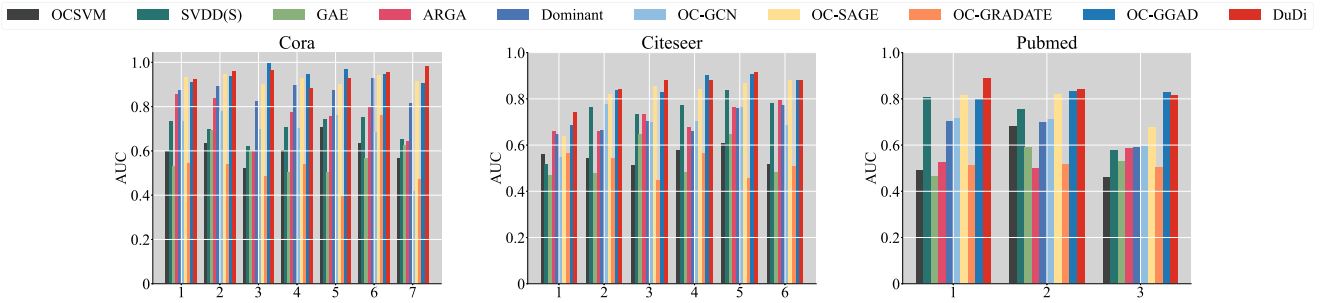


Fig. 3. Variation of Four Loss Functions During Training on the Cora Dataset.

One-class GNN based methods, including OC-GCN and OC-SAGE [21], which combine graph embeddings and one-class classification for anomaly detection. Graph SSL baselines include CCA-SSG [42], GRADATE [19], and GGAD [43]. CCA-SSG optimizes a CCA-based feature objective, GRADATE uses diverse contrastive pair sampling to capture local–global anomalous signals, and GGAD generates pseudo-anomalies using asymmetric local affinity and egocentric closeness priors in a semi-supervised setting.

We adopt the architectures of GRADATE and GGAD for one-vs-all anomaly detection tasks and named them OC-GRADATE and OC-GGAD, respectively.

C. Performance Analysis

Table II shows the results of all methods on each dataset, while Fig. 3 illustrates the performance of each method when considering each node class as the normal class on each dataset. The experimental results demonstrate that DuDi outperforms the other methods. Traditional baselines (OC-SVM, Deep-SVDD) underperform on all datasets because they model either attributes or structure in isolation, ignoring complementary information.

Autoencoder-based methods (Dominant, Conad) also underperform, as they struggle to model complex structure–attribute interactions and rely on reconstruction error, which is insensitive to subtle or non-salient anomalies.

SSL methods such as CCA-SSG perform well on smaller graphs (Cora, Citeseer) but degrade markedly on Pubmed, indicating limited scalability to complex structures. Contrastive methods (e.g., GRADATE) leverage node context via positive/negative pairs but lack explicit anomaly modeling, while GGAD’s reliance on local features and synthetic anomalies prevents it from fully capturing global dependencies, leading to suboptimal detection.

One-class graph methods, particularly OC-SAGE, outperform other baselines by more effectively exploiting structural features. DuDi further improves performance by explicitly modeling macro–micro correlation discrepancies between normal and anomalous nodes.

Fig. 3 demonstrates DuDi’s robustness: when each class is treated as normal in turn, it maintains stable performance across datasets, benefiting from explicit macro–micro correlation modeling and improved noise tolerance.

D. Ablation Study

We evaluate DuDi against three variants across datasets to examine the various components of DuDi used for anomaly detection. The variants are defined as follows:

- **wo-g**: It retains only the local-scope features and uses reconstruction error to identify anomalous nodes.

TABLE III
ABLATION STUDY. ALL VALUES ARE PERCENTAGES (%) AND AVERAGED WITH STDDEVS OVER 5 SEEDS. THE BEST RESULTS ARE MARKED IN **BOLD**

Dataset	Cora		Citeseer		Pubmed	
	AUC	AP	AUC	AP	AUC	AP
wo- <i>g</i>	49.24±3.55	50.98±3.15	49.95±3.65	50.51±2.98	50.23±0.83	49.97±0.77
wo- <i>l</i>	50.25±3.53	51.98±3.10	49.56±4.36	50.61±3.34	50.26±0.83	50.33±0.65
wo- <i>dif</i>	92.27±4.31	88.88±6.68	81.33±8.06	76.21±9.13	78.23±5.25	75.94±2.34
wo- $\mathcal{L}_{dco}$	91.58±4.15	88.08±5.58	79.50±5.69	72.44±7.49	81.76±4.05	79.65±4.06
DuDi	94.23±3.24	93.04±3.99	85.11±6.18	82.71±7.11	84.85±3.13	83.52±3.71

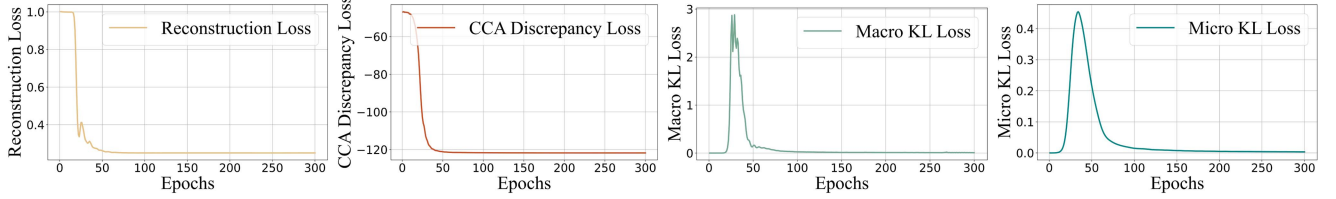


Fig. 4. Variation of Four Loss Functions During Training on the Cora Dataset.

- **wo-*l***: It maintains only the global-scope features and relies on reconstruction error to detect anomalous nodes.
- **wo-*dif***: It eliminates the graph diffusion in global-scope representation learning to assess the impact of GDC.
- **wo- $\mathcal{L}_{dco}$** : It studies the effectiveness of the decorrelation term by removing it from the loss function.

Table III presents the results that DuDi achieves the best performance across all datasets. Both wo-*g* and wo-*l* show poor performance, which is consistent with the results of some attribute network anomaly detection methods based on reconstruction error.

wo-*dif* exhibits a marked performance drop on all datasets, underscoring the importance of diffusion for capturing global structure. Pure diffusion can cause over-smoothing, blur fine-grained structural variations, and weaken the detection of subtle global dependencies. DuDi introduces a diffusion module that enhances global feature propagation while mitigating over-smoothing, thereby improving anomaly detection and alleviating the limitations of vanilla GDC in modeling global dependencies.

DuDi explicitly measures macro–micro correlation discrepancies and achieves the best results, indicating effective multi-scope representation learning. Its CCA-based discrepancy analysis simplifies anomaly pattern modeling and strengthens detection, enabling accurate and efficient anomaly recognition.

wo- $\mathcal{L}_{dco}$ performs poorly. Removing the decorrelation term hinders suppression of redundant correlations, encourages trivial solutions, and reduces overall effectiveness.

Additionally, we present the variation of four training loss functions during the training process on the Cora dataset, specifically CCA Discrepancy Loss, Reconstruction Loss, Macro KL Loss, and Micro KL Loss, as illustrated in Fig. 4. As shown in the figure, all four losses exhibit a rapid decline during the first 50 epochs, followed by a phase of stabilization. This pattern suggests that these loss functions work synergistically, rather than in conflict.

E. Parameter Sensitive Analysis

We explore the effects of hyper-parameter λ , dimensions of node embeddings, and the number of layers in the weight-sharing encoder on the performance. The results on Cora dataset are summarized in Fig. 5.

- The experimental results demonstrate that the AUC and AP exhibit relative stability across varying values of λ , with a minor decline observed around $\lambda = 10^{-2}$. Overall, the performance of DuDi remains less affected by variations in λ .
- As for the embeddings dimensions, the performance reaches equilibrium at dimension 64 with no significant gains from higher dimensions — suggesting intrinsic mechanisms preventing overfitting while preserving structural semantics. This stability reflects the effective integration of graph representation patterns from dual scopes, allowing DuDi to capture essential network structure and node attribute information, regardless of dimensionality.
- We also notice that DuDi’s performance remains stable across different layer counts. Optimal detection occurs at 3 layers where residual aggregation balances neighborhood depths. Overall, DuDi performs optimally with a moderate number of layers, beyond which additional layers do not yield significant improvements.

F. Visualization

Fig. 6 shows t-SNE visualizations of the 2D embeddings on Cora for OC-GCN, CCA-SSG, OC-GGAD, and DuDi. OC-GCN achieves partial separation between normal and abnormal nodes but degrades as the data become more complex, while CCA-SSG and OC-GGAD produce heavily mixed embeddings, indicating weak anomaly discrimination. DuDi yields the clearest, near-linear separation, suggesting stronger anomaly detection and better preservation of both local and global structures.

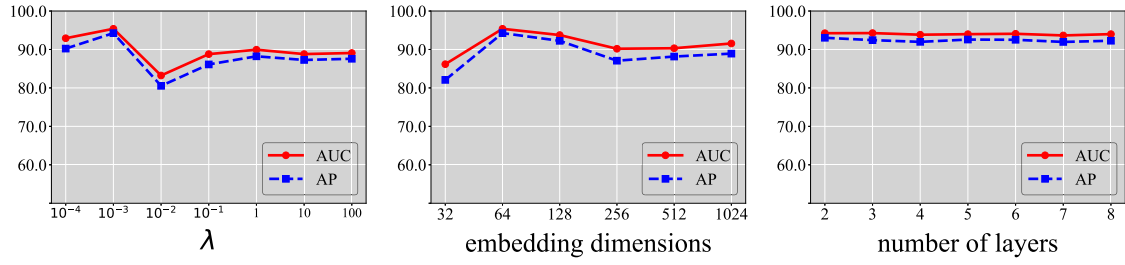


Fig. 5. The impact of different trade-off weights.

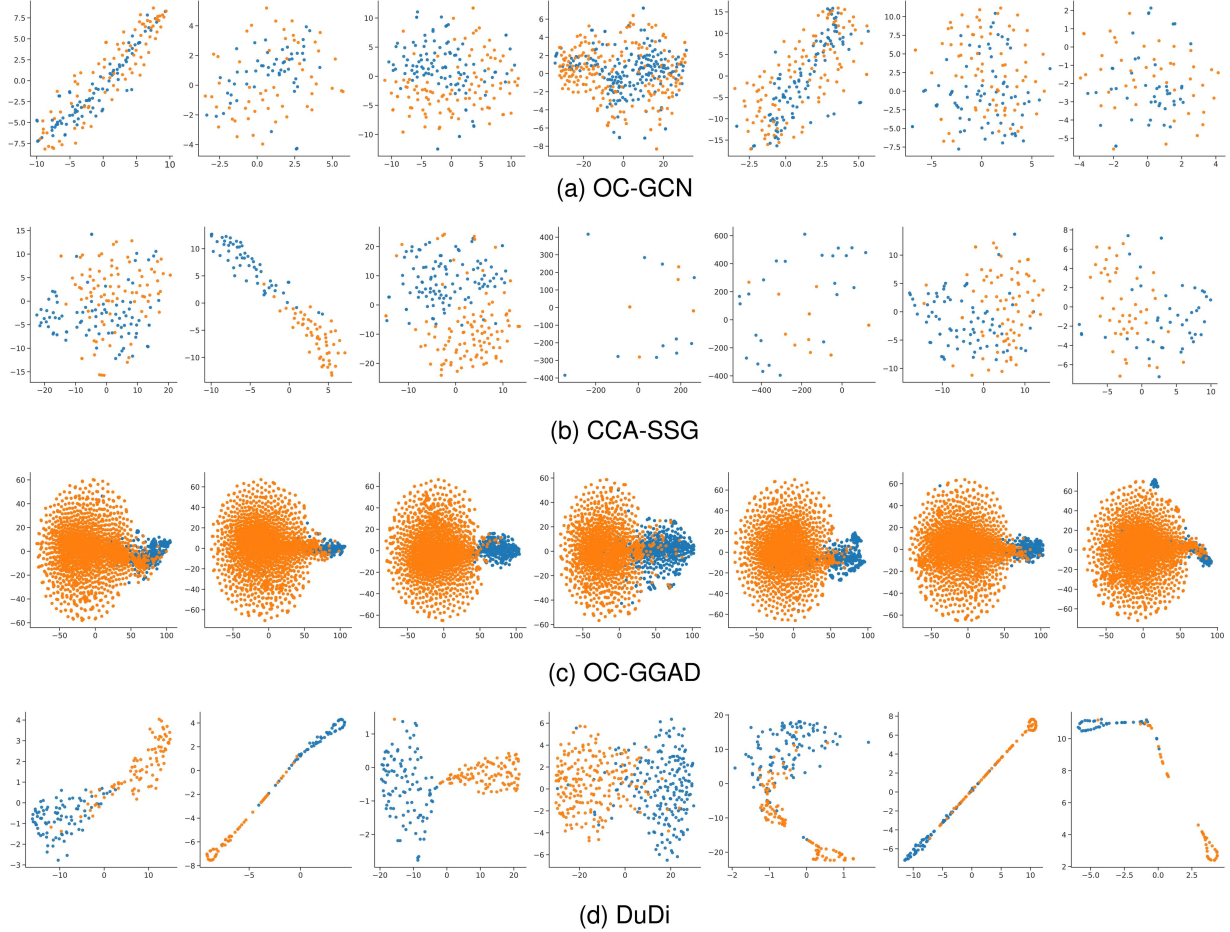


Fig. 6. T-SNE visualization on Cora dataset. (Blue points indicate the normal nodes and orange points indicate the abnormal nodes respectively.)

G. Computational Analysis

We compare the complexity of DuDi with other baselines, as shown in Table IV, in terms of model size (Params/MB), computational cost (FLOPs/G), and inference time (times/s).

DuDi exhibits a moderate computational cost, with FLOPs/G reaching 0.6584, positioning it as a balanced approach between efficiency and expressive power. Compared to OC-SAGE (4986.2413 FLOPs/G), DuDi maintains a significantly lower computational burden while still ensuring effective anomaly detection.

In terms of model size, DuDi's parameter footprint of 2.4972 MB is considerably more compact than OC-SAGE (102.6528 MB) and Dominant (3.8973 MB), highlighting its

TABLE IV
COMPUTATION ANALYSIS FOR DuDi WITH OTHER METHODS

Methods	Params/MB	FLOPs/G	times/s
Deep-SVDD(Attr) [40]	1.5669	11.8669	0.001
Deep-SVDD(Stru) [40]	1.5666	0.0024	0.005
Dominant [7]	3.8973	1.0058	0.0142
Conad [41]	1.9425	0.4951	0.0113
OC-GCN [21]	1.3046	0.4312	0.005
OC-SAGE [21]	102.6528	4986.2413	0.135
CCA-SSG [42]	0.2080	0.3111	0.106
OC-GRADATE [19]	0.1634	0.1834	0.0130
OC-GGAD [43]	1.7516	0.6662	0.0435
DuDi	2.4972	0.6584	0.0169

ability to retain model complexity without excessive memory overhead.

Regarding inference efficiency, DuDi achieves an inference time of 0.0169 seconds, aligning closely with other efficient models like OC-GCN (0.0113 s) and OC-GGAD (0.0168 s). This efficiency is attributed to DuDi's optimized graph representation learning, which effectively balances computational demand and detection accuracy, making it a practical choice for attributed network anomaly detection scenarios.

The complexity of the DuDi is dominated by the graph convolution layers. For each graph convolution layer, the complexity is $O(N \cdot d \cdot h)$, where N is the number of nodes, d is the average degree of nodes, and h is the number of attention heads. With L layers, the overall complexity becomes $O(L \cdot N \cdot d \cdot h)$. For the total loss, the main computational cost comes from matrix multiplications, which have a complexity of $O(N^2)$, as shown in (10), (11), and (8). The total complexity of the loss function is thus $O(N^2)$.

V. CONCLUSION

This work addresses the critical yet underexplored dual-scope correlation discrepancy principle in attributed network anomaly detection: normal nodes exhibit strong consistency between macro (global) and micro (local) scopes, while anomalies display inherent discrepancies. To operationalize this discriminative property, we propose DuDi, a novel framework that (1) amplifies anomaly discrimination through macro-micro scope discrepancy maximization, achieved by GDC-based macro pattern denoising (suppressing spurious edges/attribute noise) and localized micro pattern sampling, and (2) quantifies anomaly saliency via CCA-driven discrepancy scoring. The superiority of DuDi, validated across multiple real-world datasets, establishes a new paradigm for detecting anomalies through dual-scope correlation dynamics rather than conventional representation learning.

In the future, we aim to improve DuDi's generalization capability, and consider the integration of multiple data sources and different types of graph structures to build a more comprehensive and powerful anomaly detection system.

REFERENCES

- [1] M. Jin et al., "A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10466–10485, Dec. 2024.
- [2] S. Xiang et al., "Semi-supervised credit card fraud detection via attribute-driven graph representation," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 12, pp. 14557–14565.
- [3] W. Ju et al., "A comprehensive survey on deep graph representation learning," *Neural Netw.*, vol. 173, 2024, Art. no. 106207.
- [4] B. Wang and B. Jiang, "Data representation via attribute selection- propagation neural network," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 9, pp. 258–267, 2023.
- [5] S. Zhang, H. Tong, J. Xu, and R. Maciejewski, "Graph convolutional networks: A comprehensive review," *Comput. Social Netw.*, vol. 6, no. 1, pp. 1–23, 2019.
- [6] J. Tang, J. Li, Z. Gao, and J. Li, "Rethinking graph neural networks for anomaly detection," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 21076–21089.
- [7] K. Ding, J. Li, R. Bhanushali, and H. Liu, "Deep anomaly detection on attributed networks," in *Proc. SIAM Int. Conf. Data Mining*, 2019, pp. 594–602.
- [8] Y. Liu, Z. Li, S. Pan, C. Gong, C. Zhou, and G. Karypis, "Anomaly detection on attributed networks via contrastive self-supervised learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2378–2392, Jun. 2022.
- [9] H. Qiao, H. Tong, B. An, I. King, C. Aggarwal, and G. Pang, "Deep graph anomaly detection: A survey and new perspectives," 2024, doi: [10.1109/TKDE.2025.3581578](https://doi.org/10.1109/TKDE.2025.3581578).
- [10] A. Ahmed, N. Shervashidze, S. Narayanamurthy, V. Josifovski, and A. J. Smola, "Distributed large-scale natural graph factorization," in *Proc. 22nd Int. Conf. World Wide Web*, 2013, pp. 37–48.
- [11] B. Perozzi, R. Al-Rfou, and S. Skiena, "DeepWalk: Online learning of social representations," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2014, pp. 701–710.
- [12] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 855–864.
- [13] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *Int. Conf. Learning Representations (ICLR)*, pp. 1–14, 2017. [Online]. Available: <https://openreview.net/forum?id=SJU4ayYgl>.
- [14] J. Gasteiger, S. Weißenberger, and S. Günnemann, "Diffusion improves graph learning," in *Proc. Adv. Neural Inf. Process. Syst.*, pp. 13298–13310, 2019, vol. 32.
- [15] J. H. Giraldo, K. Skianis, T. Bouwmans, and F. D. Malliaros, "On the trade-off between over-smoothing and over-squashing in deep graph neural networks," in *Proc. 32nd ACM Int. Conf. Inf. Knowl. Manage.*, 2023, pp. 566–576.
- [16] O. A. Ekle and W. Eberle, "Anomaly detection in dynamic graphs: A comprehensive survey," *ACM Trans. Knowl. Discov. Data*, vol. 18, no. 8, pp. 1–44, Jul. 2024.
- [17] H. Fan, F. Zhang, and Z. Li, "Anomalydae: Dual autoencoder for anomaly detection on attributed networks," in *Proc. ICASSP 2020-2020 IEEE Int. Conf. Acoust., Speech Signal Process.*, 2020, pp. 5685–5689.
- [18] M. Jin, Y. Liu, Y. Zheng, L. Chi, Y.-F. Li, and S. Pan, "ANEMONE: Graph anomaly detection with multi-scale contrastive learning," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manage.*, 2021, pp. 3122–3126.
- [19] J. Duan et al., "Graph anomaly detection via multi-scale contrastive learning networks with augmented view," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 6, pp. 7459–7467.
- [20] H. Qiao and G. Pang, "Truncated affinity maximization: One-class homophily modeling for graph anomaly detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, vol. 36.
- [21] X. Wang, B. Jin, Y. Du, P. Cui, Y. Tan, and Y. Yang, "One-class graph neural networks for anomaly detection in attributed networks," *Neural Comput. Appl.*, vol. 33, pp. 12073–12085, 2021.
- [22] F. Zhang, H. Fan, R. Wang, Z. Li, and T. Liang, "Deep dual support vector data description for anomaly detection on attributed networks," *Int. J. Intell. Syst.*, vol. 37, no. 2, pp. 1509–1528, 2022.
- [23] S. Zhou, X. Huang, N. Liu, Q. Tan, and F.-L. Chung, "Unseen anomaly detection on networks via multi-hypersphere learning," in *Proc. 2022 SIAM Int. Conf. Data Mining*, 2022, pp. 262–270.
- [24] X. Yuan, N. Zhou, S. Yu, H. Huang, Z. Chen, and F. Xia, "Higher-order structure based anomaly detection on attributed networks," in *Proc. 2021 IEEE Int. Conf. Big Data*, 2021, pp. 2691–2700.
- [25] H. Qiao, H. Tong, B. An, I. King, C. Aggarwal, and G. Pang, "Deep graph anomaly detection: A survey and new perspectives," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5106–5126, Sep. 2025.
- [26] C. Xiao, X. Xu, Y. Lei, K. Zhang, S. Liu, and F. Zhou, "Counterfactual graph learning for anomaly detection on attributed networks," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 10, pp. 10540–10553, Oct. 2023.
- [27] J. Duan, B. Xiao, S. Wang, H. Zhou, and X. Liu, "ARISE: Graph anomaly detection on attributed networks via substructure awareness," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 18172–18185, Dec. 2024.
- [28] M. Shao, Y. Lin, Q. Peng, J. Zhao, Z. Pei, and Y. Sun, "Learning graph deep autoencoder for anomaly detection in multi-attributed networks," *Knowl.-Based Syst.*, vol. 260, 2023, Art. no. 110084.
- [29] J. He, Q. Xu, Y. Jiang, Z. Wang, and Q. Huang, "ADA-GAD: Anomaly-denoised autoencoders for graph anomaly detection," in *Proc. 38th AAAI Conf. Artif. Intell.*, M. J. Wooldridge, J. G. Dy, and S. Natarajan, Eds., 2024, pp. 8481–8489.

- [30] M. Munir, S. A. Siddiqui, A. Dengel, and S. Ahmed, "DeepAnT: A deep learning approach for unsupervised anomaly detection in time series," *IEEE Access*, vol. 7, pp. 1991–2005, 2019.
- [31] F. Xu et al., "GLADformer: A mixed perspective for graph-level anomaly detection," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discov. Databases*, 2024, pp. 337–353.
- [32] Y. Chen, B. Coskunuzer, and Y. Gel, "Topological relational learning on graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 27029–27042.
- [33] Y. Zhang, J. Shen, R. Zhang, and Z. Zhao, "Network representation learning via improved random walk with restart," *Knowl.-Based Syst.*, vol. 263, 2023, Art. no. 110255.
- [34] S. Pan, R. Hu, G. Long, J. Jiang, L. Yao, and C. Zhang, "Adversarially regularized graph autoencoder for graph embedding," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, 2018, pp. 2609–2615.
- [35] H. Fan et al., "Heterogeneous hypergraph variational autoencoder for link prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4125–4138, Aug. 2022.
- [36] W. Fan et al., "Deep adversarial canonical correlation analysis," in *Proc. 2020 SIAM Int. Conf. Data Mining*, 2020, pp. 352–360.
- [37] Y. Xu et al., "Revisiting graph contrastive learning on anomaly detection: A structural imbalance perspective," in *Proc. 39th AAAI Conf. Artif. Intell.*, 2025, pp. 12972–12980.
- [38] R. Wang, F. Zhang, X. Huang, C. Tian, L. Xi, and H. Fan, "CaCo: Attributed network anomaly detection via canonical correlation analysis," *IEEE Trans. Ind. Informat.*, vol. 20, no. 1, pp. 461–470, Jan. 2024.
- [39] P. Nader, P. Honeine, and P. Beausery, " l_p -norms in one-class classification for intrusion detection in SCADA systems," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2308–2317, Nov. 2014.
- [40] L. Ruff et al., "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4393–4402.
- [41] Z. Xu, X. Huang, Y. Zhao, Y. Dong, and J. Li, "Contrastive attributed network anomaly detection with data augmentation," in *Proc. Pacific-Asia Conf. Knowl. Discov. Data Mining*, 2022, pp. 444–457.
- [42] H. Zhang, Q. Wu, J. Yan, D. Wipf, and P. S. Yu, "From canonical correlation analysis to self-supervised graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 76–89.
- [43] H. Qiao, Q. Wen, X. Li, E.-P. Lim, and G. Pang, "Generative semi-supervised graph anomaly detection," *Adv. Neural Inf. Process. Syst.*, pp. 4660–4688, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/file/085b4b5d1f81ad9e057ad2b3de922ad4-Paper-Conference.pdf.



Liang Xi received the Ph.D. degree in computer applied technology from the Harbin University of Science and Technology, Harbin, China, in 2012. He was with The University of Texas at Dallas, Richardson, TX, USA, as a Visiting Scholar. He is currently a Professor with the Harbin University of Science and Technology, Harbin. His research interests include data mining, machine learning, pattern recognition, and network security.



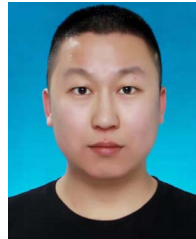
Runze Li received the M.S. degree in computer science and technology in 2025 from the Harbin University of Science and Technology, Harbin, China, where he is currently working toward the Ph.D. degree. His research interests include artificial intelligence, network security, and machine learning.



Dichen Li is currently working toward the master's degree in computer science and technology with the Harbin University of Science and Technology, Harbin, China. His research interests include graph representation learning and anomaly detection.



Ruidong Wang received the M.S. degree in computer science and technology in 2020 from the Harbin University of Science and Technology, Harbin, China, where he is currently working toward the Ph.D. degree. His research interests include graph data mining, time series analysis, and anomaly detection.



Han Liu received the M.S. degree in computer science and technology in 2021 from the Harbin University of Science and Technology, Harbin, China, where he is currently working toward the Ph.D. degree. His research interests include artificial intelligence, network security, and machine learning.



Huiying Xu received the M.S. degree from the National University of Defense Technology, Changsha, China. She is currently an Associate Professor with the School of Computer Science and Technology, Zhejiang Normal University, Jinhua, China, and also a Researcher with the Research Institute of Ningbo Cixing Company Ltd., China. She is a member of the China Computer Federation.



Xinzhong Zhu (Member, IEEE) received the M.S. degree from the National University of Defense Technology, Changsha, China, and the Ph.D. degree from Xidian University, Xi'an, China. He is currently a Professor with the School of Computer Science and Technology, Zhejiang Normal University, Jinhua, China, the Chief Scientist with Beijing Geekplus Technology Company, Ltd., and the President with the Research Institute of Ningbo Cixing Company, Ltd., China.